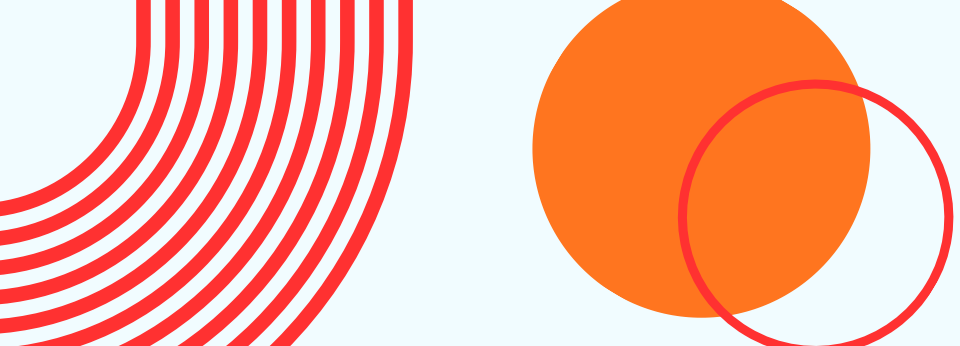




Zainab Lorgat



CVSS v4

Open framework used to assess and
communicate the severity of software
vulnerabilities



Attack Vector

Where can the vulnerability be exploited from?

Privileges Required

Were any privileges needed to exploit the vulnerability?

Attack Requirements

Was the attack dependent on a specific configuration in order to be exploited?

Attack Complexity

How difficult was it to conduct the attack?

User Interaction

Did the user have to interact with anything for the attack to be conducted?

Base Metrics

METRIC	DESCRIPTION
Attack Vector (AV)	Defines how the attack is delivered (e.g., Network, Adjacent, Local, Physical).
Attack Complexity (AC)	Indicates the complexity of the attack required (e.g., Low, High).
Attack Requirements (AT)	Specifies any special conditions that must be met for exploitation.
Privileges Required (PR)	Details the level of access needed to exploit the vulnerability (e.g., None, Low, High).
User Interaction (UI)	Indicates whether exploiting the vulnerability requires user action (e.g., None, Required).

Vulnerability System Impact Metrics



Confidentiality

Is sensitive information accessible by unauthorized users?



Integrity

Can information be modified? How consistent and accurate is the data?



Availability

Are resources available to their respective users as they should be?

Measure the impact of each on both system and subsystem



Supplemental Environmental and Threat Metrics

METRIC	DESCRIPTION
Safety (S)	Indicates if the vulnerability poses a safety risk.
Automatable (AU)	Specifies if the exploit can be automated.
Recovery (R)	Evaluates how easy it is to recover from the exploitation.
Value Density (V)	Reflects how valuable the targeted assets are.
Vulnerability Response Effort (RE)	Assesses the effort required to respond to the vulnerability.
Provider Urgency (U)	Indicates how quickly the vendor should address the issue.

Supplemental Metrics

These metrics provide further context regarding the vulnerability.

Exploit Maturity (E)

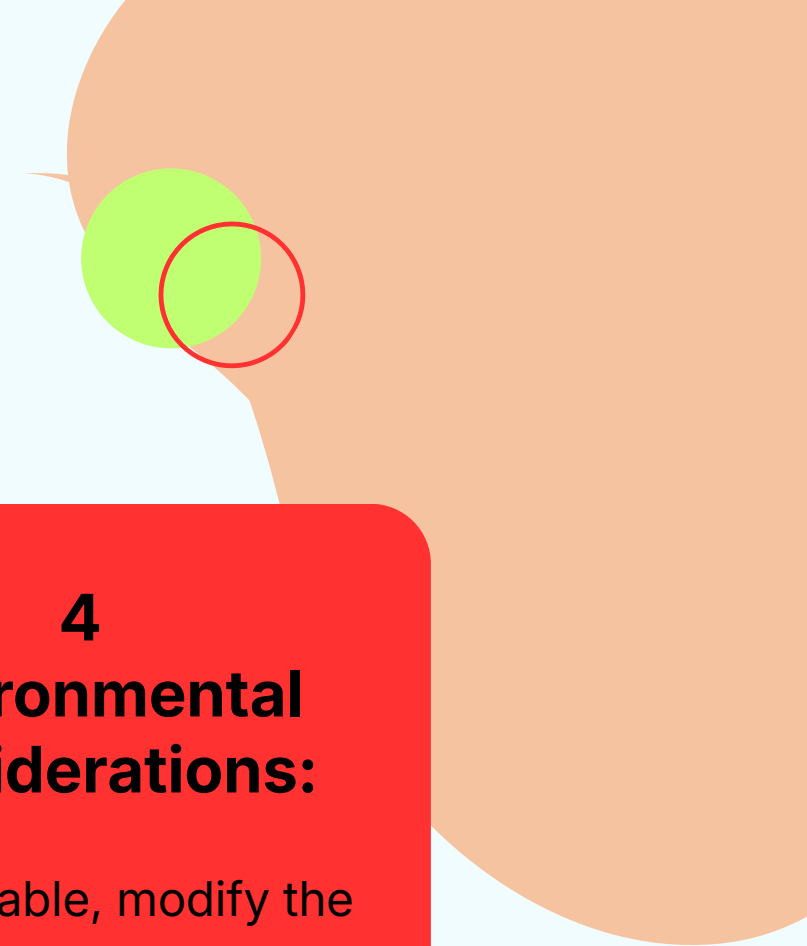
Indicates the maturity of known exploit code for the vulnerability.

Threat Metric

METRIC	DESCRIPTION
Exploitability Metrics	Adjusted metrics for exploitability based on the organization's context.
Confidentiality Requirements (CR)	Customizes confidentiality importance in the specific environment.
Integrity Requirements (IR)	Customizes integrity importance in the specific environment.
Availability Requirements (AR)	Customizes availability importance in the specific environment.

Environmental Metrics

These metrics adapt the base metrics to the specific environment.



1 Assess Base Metrics:

Determine the values for each Base Metric

2 Calculate Base Score:

Use the CVSS calculator to combine metrics into a Base Score ranging from 0.0 to 10.0.

3 Adjust with Other Metrics:

Apply any supplemental, Threat and Environmental Metric values to refine the score.

4 Environmental Considerations:

If applicable, modify the score further based on Environmental Metrics.



The formula includes a combination of factors from Base Metrics, resulting in a score between 0.0 (no severity) to 10.0 (critical).

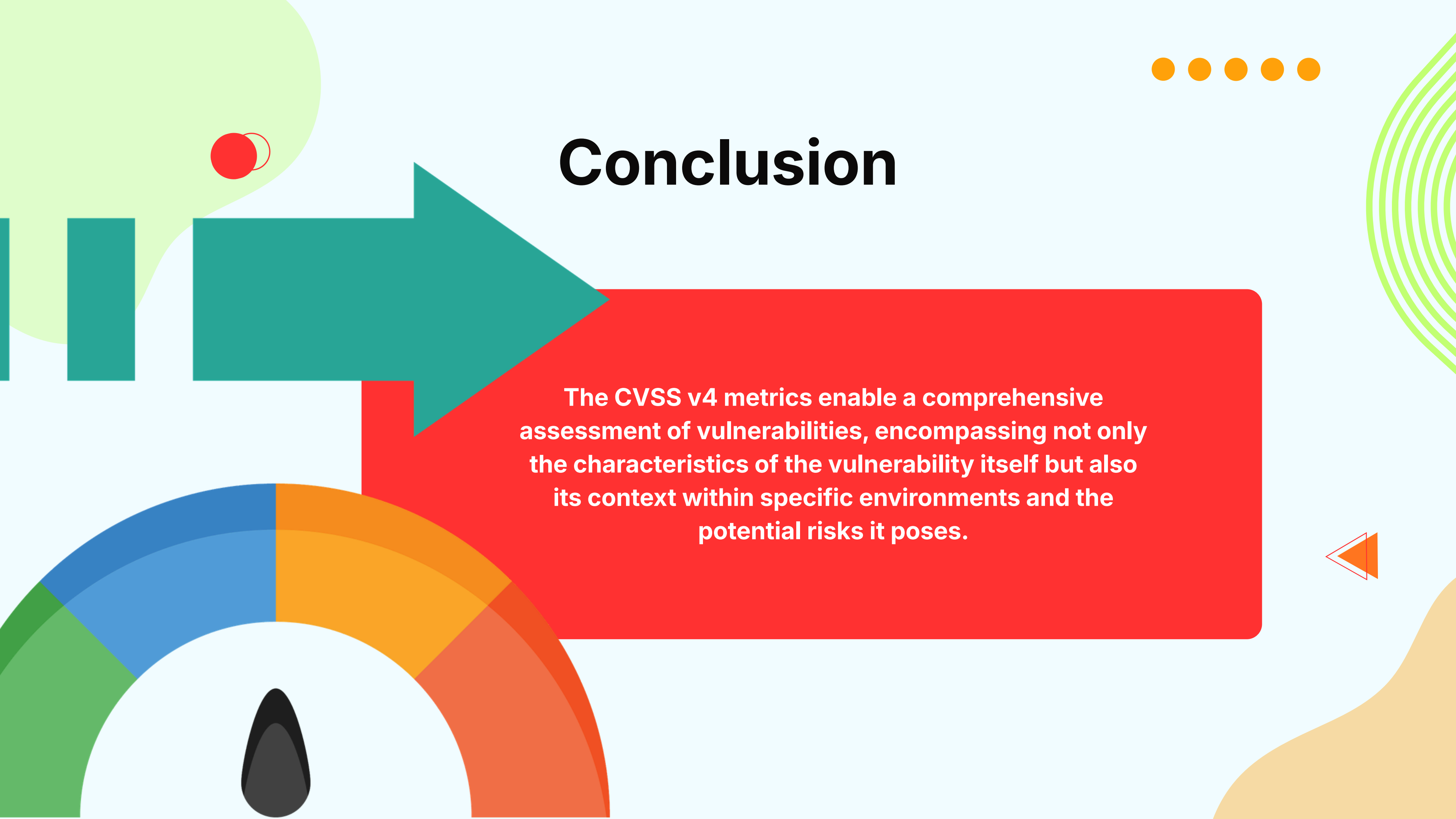


Calculating CVSS Score



Risk Ratings

SCORE RANGE	SEVERITY LEVEL	INTERPRETATION
0.0 - 3.9	Low	Low vulnerability requires little attention.
4.0 - 6.9	Medium	Moderate vulnerability may require some attention.
7.0 - 8.9	High	High vulnerability requires immediate attention and remediation.
9.0 - 10.0	Critical	Critical vulnerability requires urgent action to mitigate risk.



Conclusion

The CVSS v4 metrics enable a comprehensive assessment of vulnerabilities, encompassing not only the characteristics of the vulnerability itself but also its context within specific environments and the potential risks it poses.